Joshua Hu

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Education

University of Maryland | Bachelor of Information Science | College Park, MD May 2025

Technical Skills

Languages: Python, SQL, Java, TypeScript, JavaScript, C#, C++, C, HTML/CSS
Cloud and DevOps: AWS (Lambda, API Gateway, SQS, ECS, S3, CloudWatch), Crossplane, Harness CI/CD, Docker
Data and Search: OpenSearch (HNSW vector search), PostgreSQL, DynamoDB, RDS
Frameworks: React, Node.js, Spring Boot, .NET, Vue.js, Angular, OIDC/PKCE
Clearance: Public Trust

Experience

Software Engineer | SAIC | Reston, VA July 2025 - Present

- Engineered a serverless 1:N vector search platform on AWS (~40 Lambda functions and SQS queues) replacing a legacy Kubernetes-based system, increasing throughput **10x to 200+ requests/sec** at **sub-second p90 latency**
- Scaled Amazon OpenSearch HNSW indexes to **~3 billion** multimodal biometric embeddings (face, iris, fingerprint) under continuous real-time enrollment and large-scale backfill ingests
- Benchmarked HNSW and cluster parameters (ef_search, ef_construction, sharding, node topology) with the AWS OpenSearch team, meeting all mandated recall and false-match accuracy targets plus internal latency and concurrency goals
- Integrated multiple third-party ML embedding vendors into a vendor-agnostic extraction pipeline; designed storage across RDS, DynamoDB, and S3 for identity and travel-document metadata
- Lead a 2-engineer sub-team building a containerized React demo app on ECS used to showcase the platform to prospective clients, including PIV smart-card login via OIDC authorization code flow with PKCE
- Provision AWS infrastructure as code with Crossplane and build Harness CI/CD pipelines for multi-environment deployments; mentor junior developers on a 25-person engineering team

Software Engineer Intern | SAIC | Reston, VA July 2024 - June 2025

- Spearheaded the overhaul of a deprecated tool cataloging activities across the Department of Homeland Security into a full-stack app with an interactive analytics dashboard, cutting a manual analyst workflow from **hours to near-instant** (Vue.js, C#, .NET, PostgreSQL, AWS)
- Authored API endpoints and data-cleaning pipelines feeding an internal search engine (C#, Python, SQL)
- Owned daily ingestion of **~20,000 records** from multiple source systems into the core relational database (SQL, RDS)

Software Engineer Intern | Wangend LLC | Rockville, MD June 2023 - August 2023

- Built an ordering and modification system for a customized meal delivery service (TypeScript, Java, SQL)
- Migrated the frontend from MDBootstrap to Angular Material and replaced Stripe with Cardknox to **lower payment processing fees** (Angular, TypeScript, Java)
- Hardened backend error and edge-case handling, **reducing checkout errors by ~15%** (Spring, Java)

Projects

MC Server Bot | Python, AWS (EC2, Lambda), Discord.js, React.js May 2024

- Built a Discord bot that controls an EC2-hosted Minecraft server by invoking Lambda functions from chat commands, used by a **20+ player community**
- Cut hosting costs by **~70%** by starting the instance on demand instead of running it 24/7; added a React portal for authorized users to manage the server